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TASK	WIFI HACKING 101 REPORT

Contents

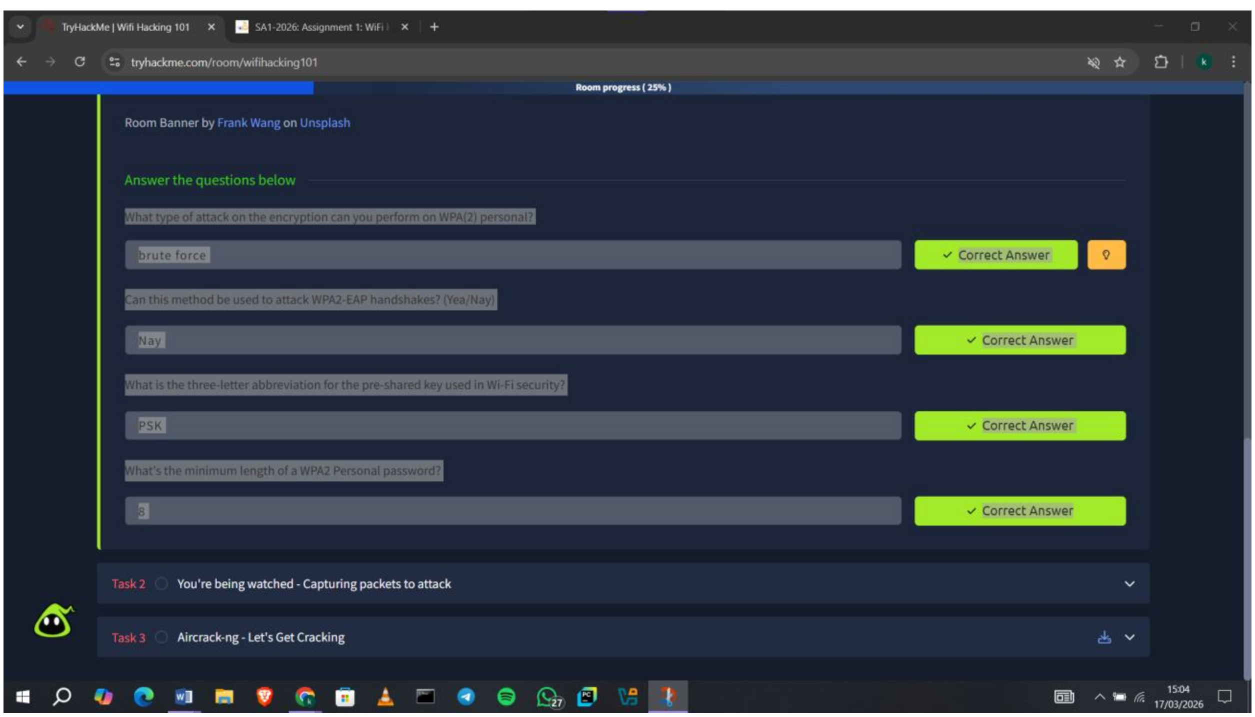
WIFI HACKING 101 REPORT

Introduction

In this module, I learned key WiFi concepts and authentication methods used in modern networks. I also gained hands-on experience using tools to capture packets and begin attacking WPA/WPA2 networks.

1.0 The Basics

I learned the key WiFi terms like SSID, BSSID, and the difference between WPA2-PSK and WPA2-EAP authentication. I also understood how the 4-way handshake works to verify devices without sharing the actual password. Additionally, I saw why WEP is insecure and how WPA2 security relies on both the password and network name, making attacks harder.

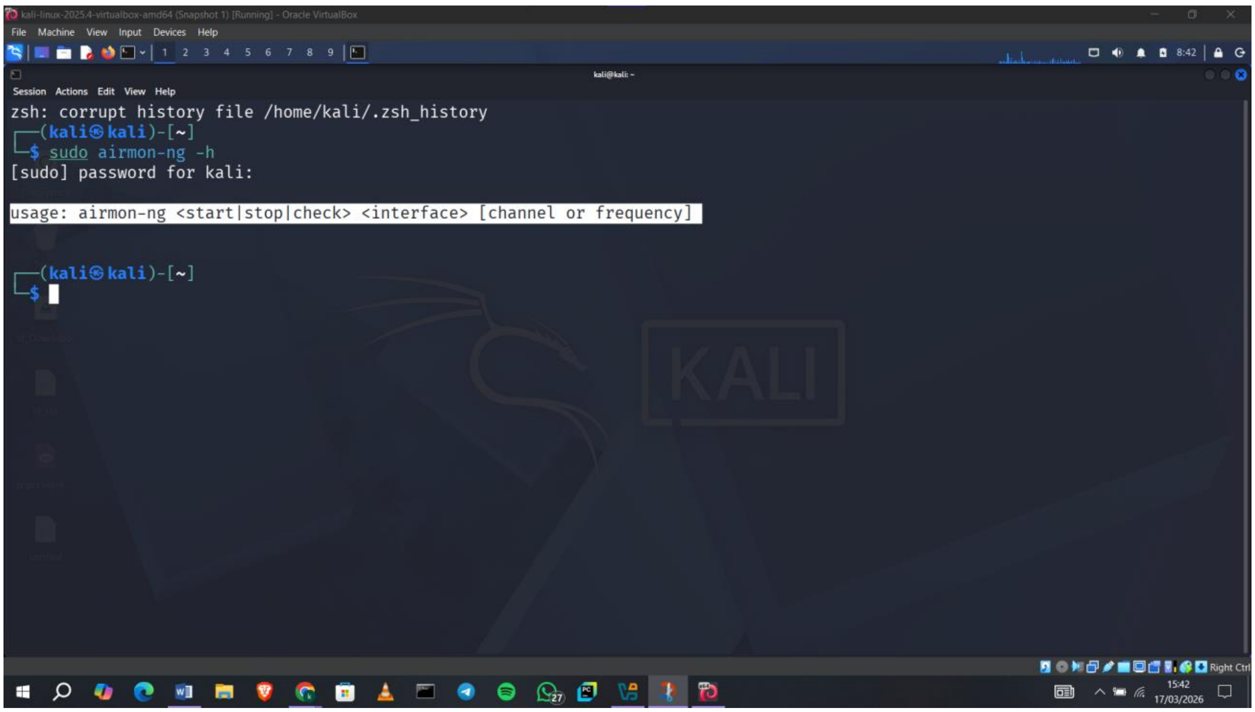


2.0 Capturing Packets to attack

I learned how to use the Aircrack-ng suite, especially tools like airmon-ng, airodump-ng, and aircrack-ng, to begin attacking WPA networks.

Question1: How do you put the interface "wlan0" into monitor mode with Aircrack tools? (Full command)

Answer: **airmon-ng start wlan0**

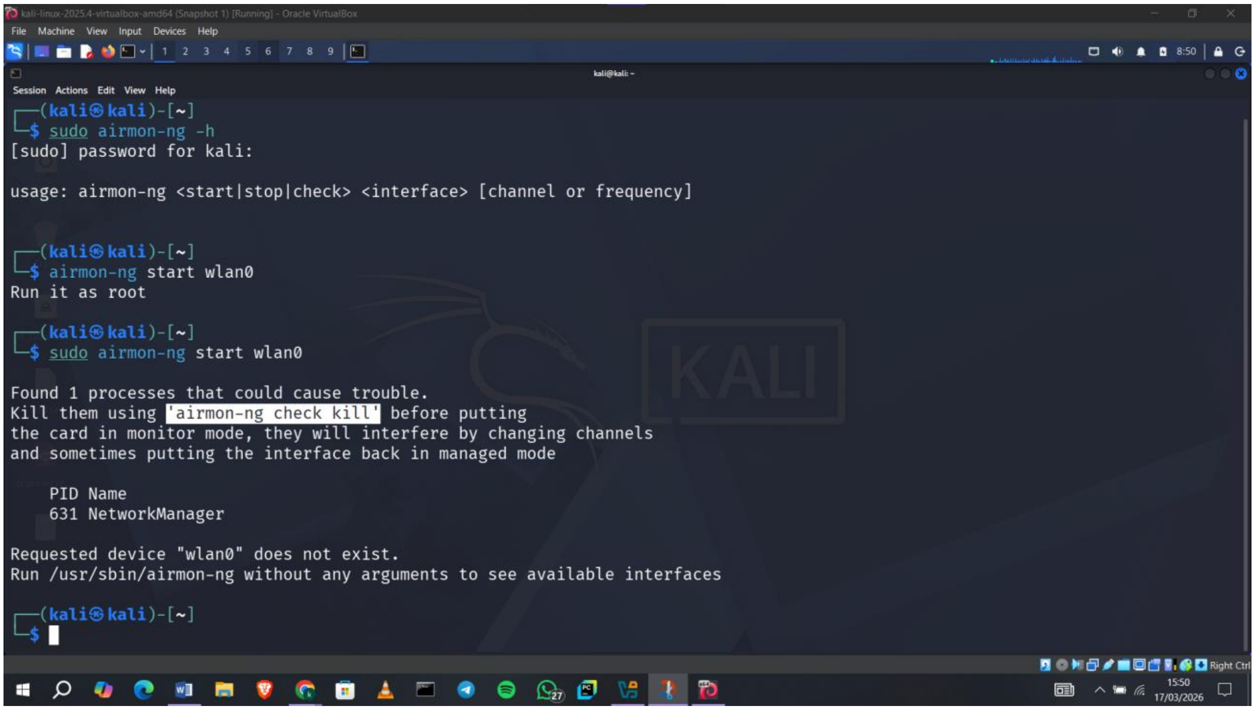


Question2: What is the new interface name likely to be after you enable monitor mode?

Answer: **wla0mon**

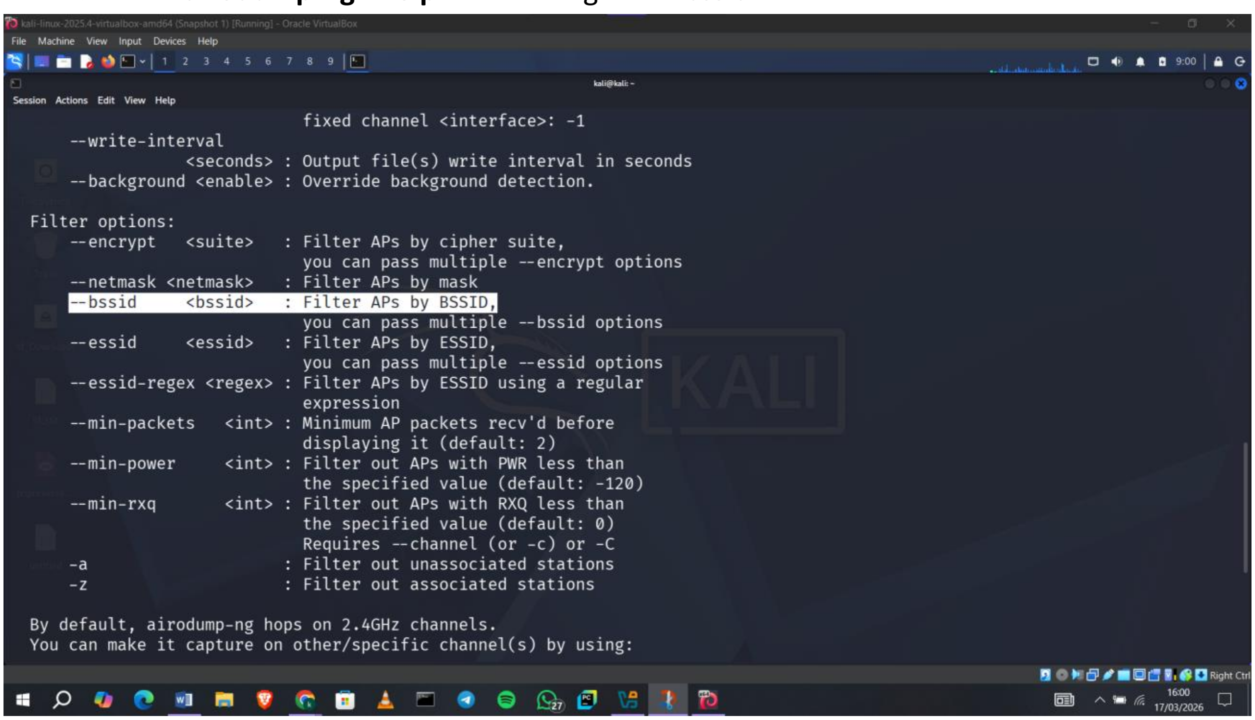
Question3: What do you do if other processes are currently trying to use that network adapter?

Answer: **airmon-ng check kill**



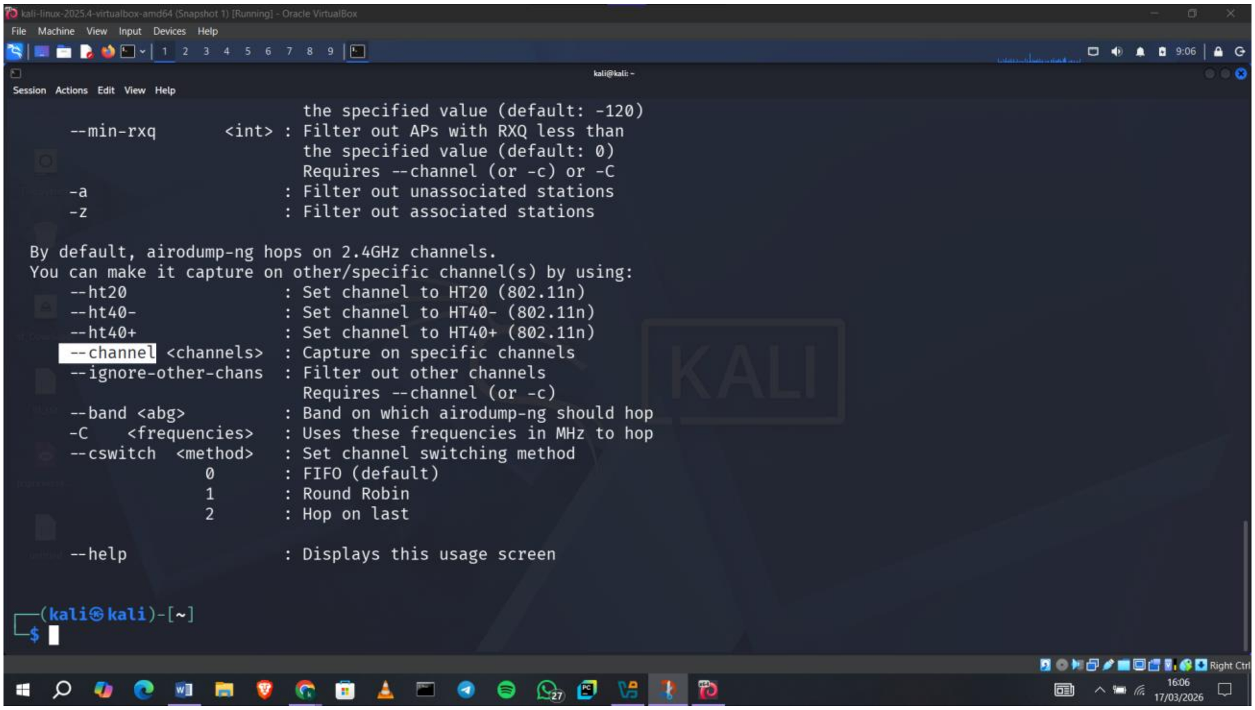
Question4: What flag do you use to set the BSSID to monitor?

Answer: I ran **airodump-ng –help** and the flag was **--bssid**



Question5: And to set the channel?

Answer: **--channel**



Question6: And how do you tell it to capture packets to a file?

Answer: **-w**

3.0 Aircrack-ng

Question1: What flag do we use to specify a BSSID to attack?

Answer: **-b**

Question2: What flag do we use to specify a wordlist?

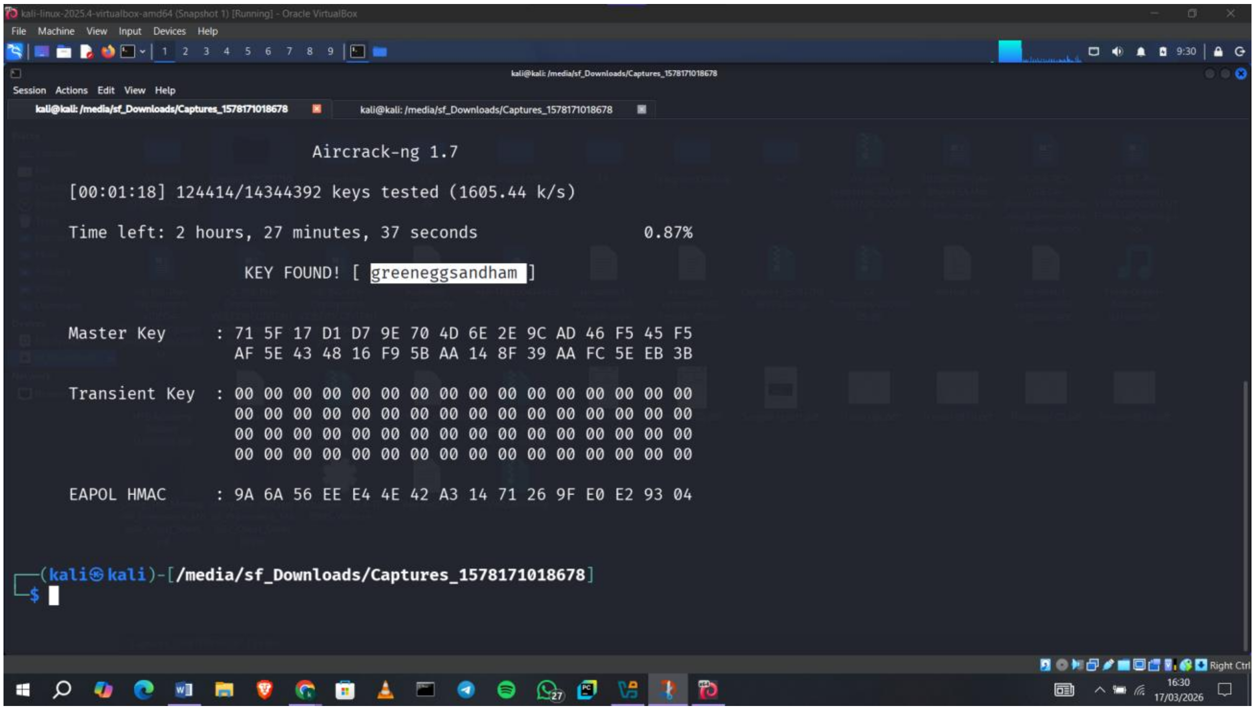
Answer: **-w**

Question3: How do we create a HCCAPX in order to use hashcat to crack the password?

Answer: **-j**

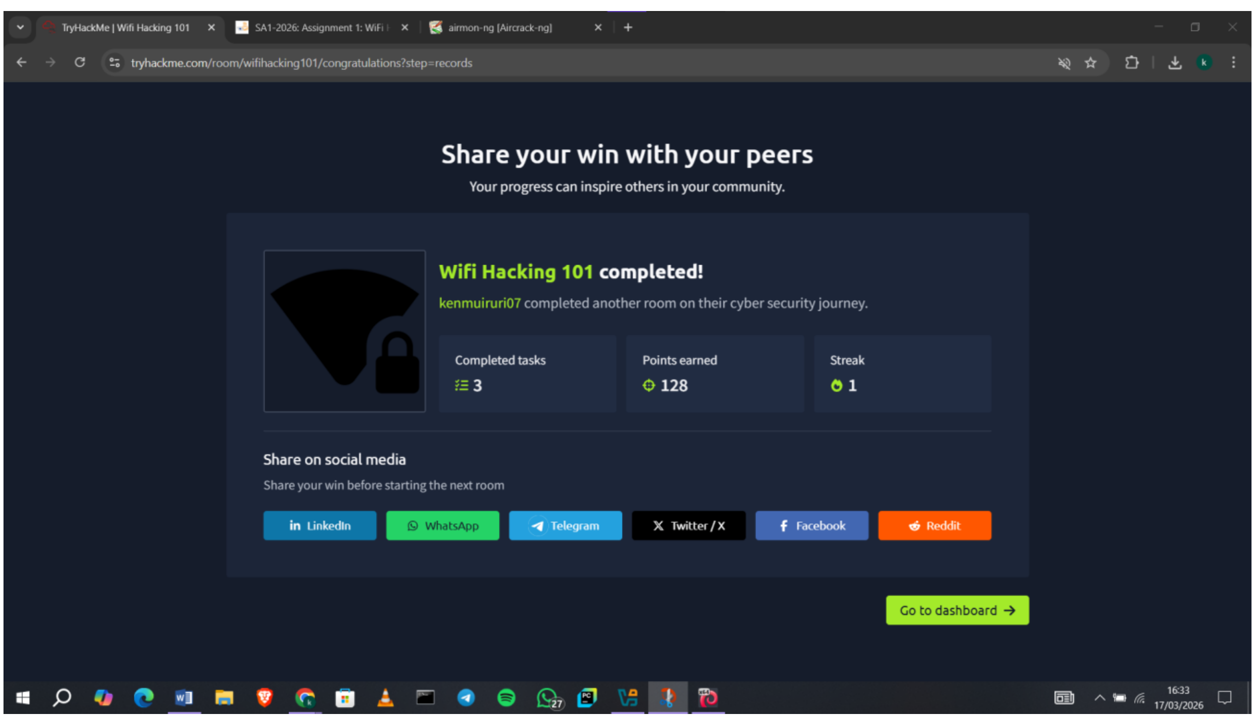
Question4: Using the rockyou wordlist, crack the password in the attached capture. What's the password?

Answer: I ran **sudo aircrack-ng -b 02:1A:11:FF:D9:BD -w /usr/share/wordlists/rockyou.txt NinjaJc01-01.cap** and found the password as **greeneggsandham**



Question5: Where is password cracking likely to be fastest, CPU or GPU?

Answer: **GPU**



Conclusion

Overall, this module helped me build a practical understanding of WiFi attacks, especially capturing and cracking WPA/WPA2 handshakes.